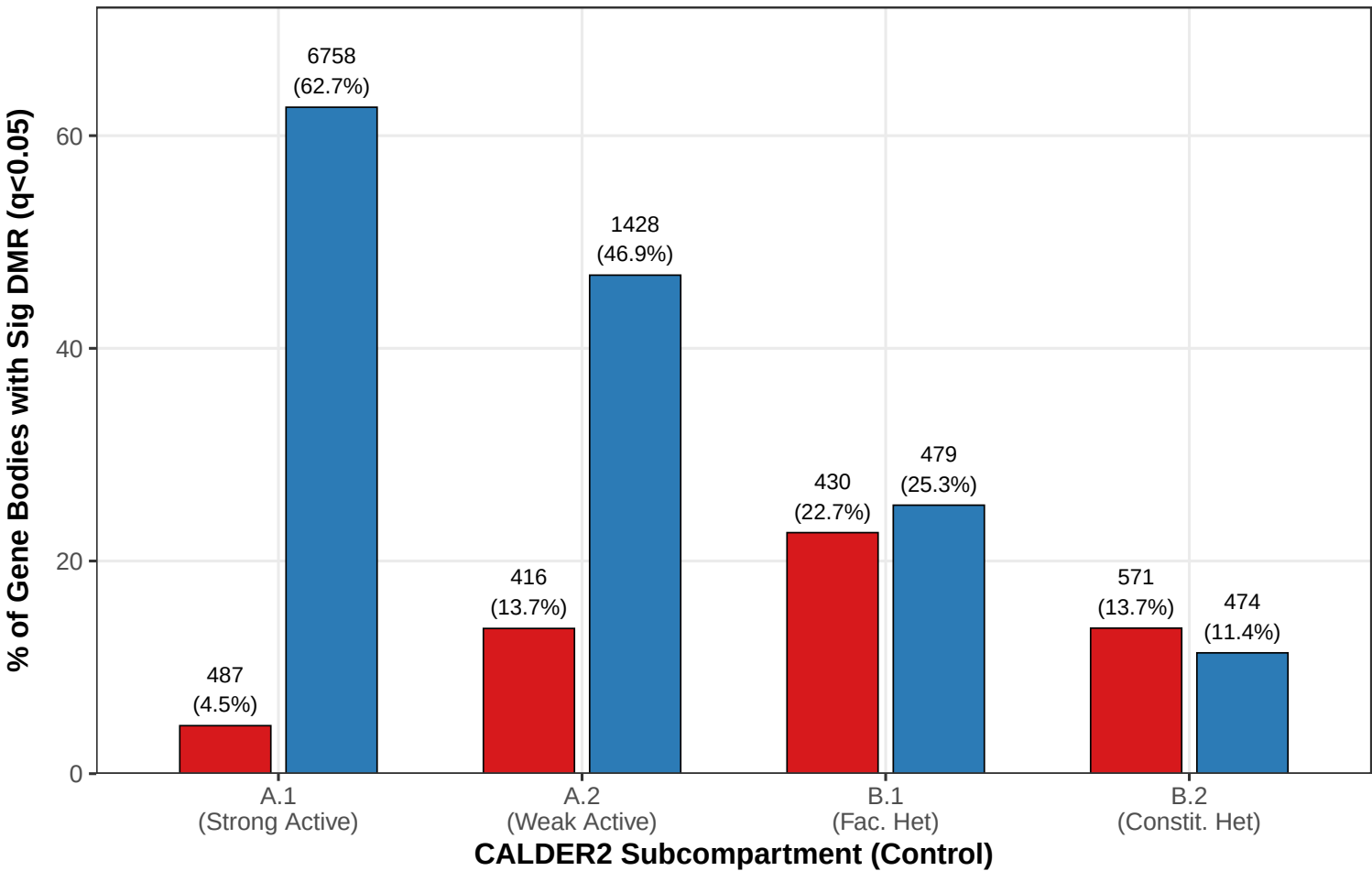


5hmC Subcompartment Analysis + Gradient Comparison

Significant 5hmC DMRs by Subcompartment

Chi² p < 2.2e-16 | Compare with 5mC gradient in Section 66a

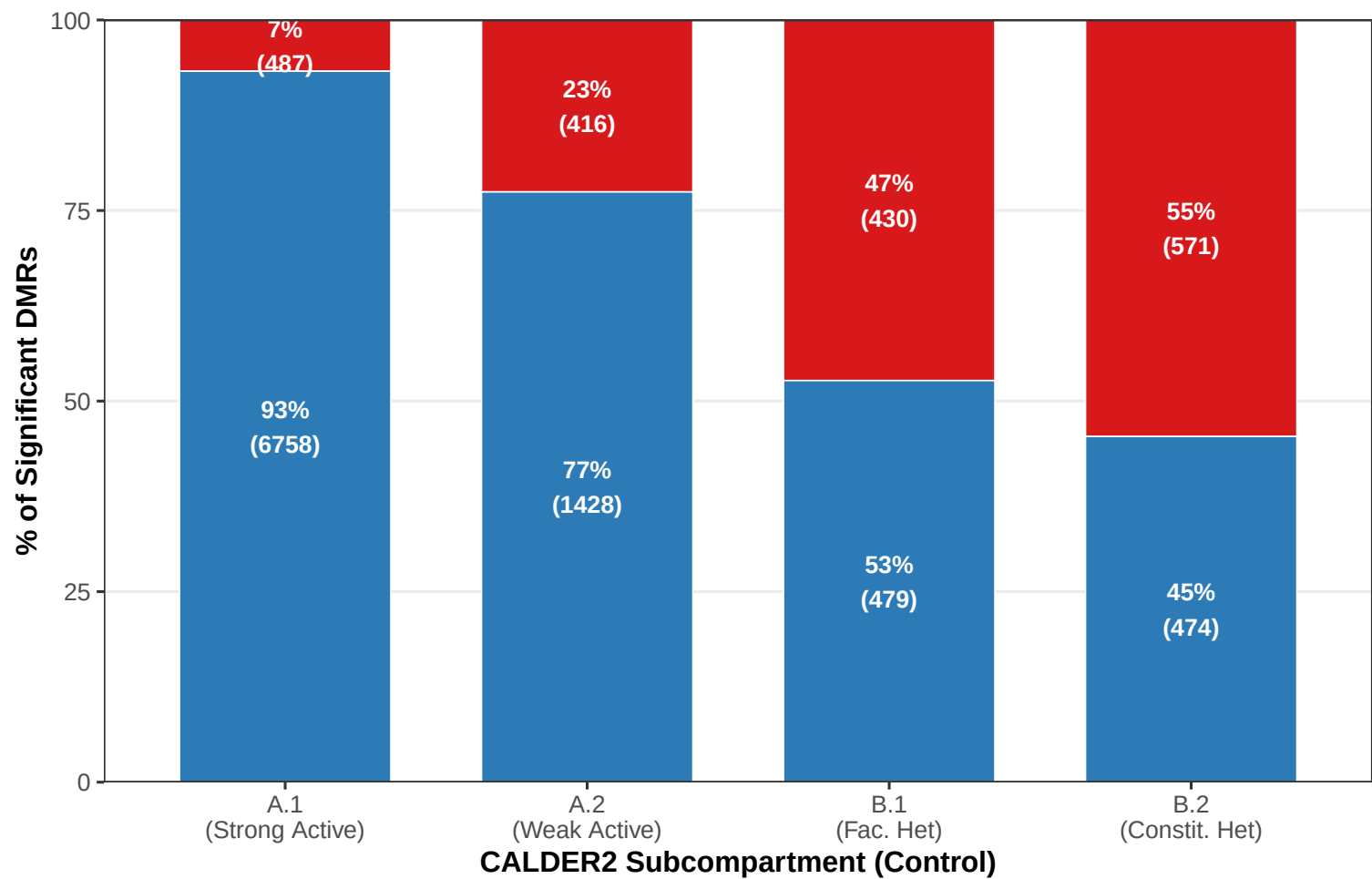
Direction ■ Hypermethylated ■ Hypomethylated



5hmC Change Direction by Subcompartment

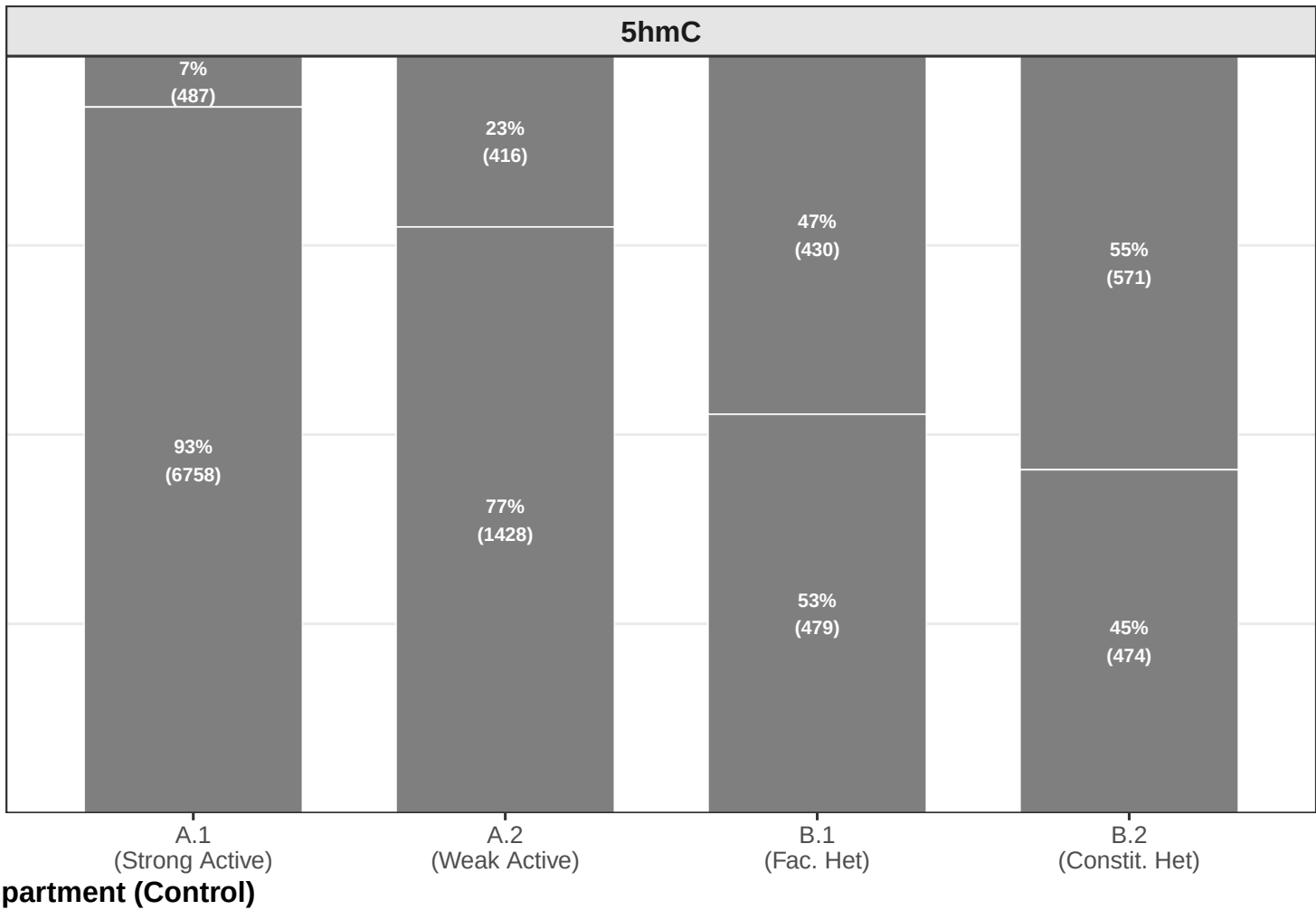
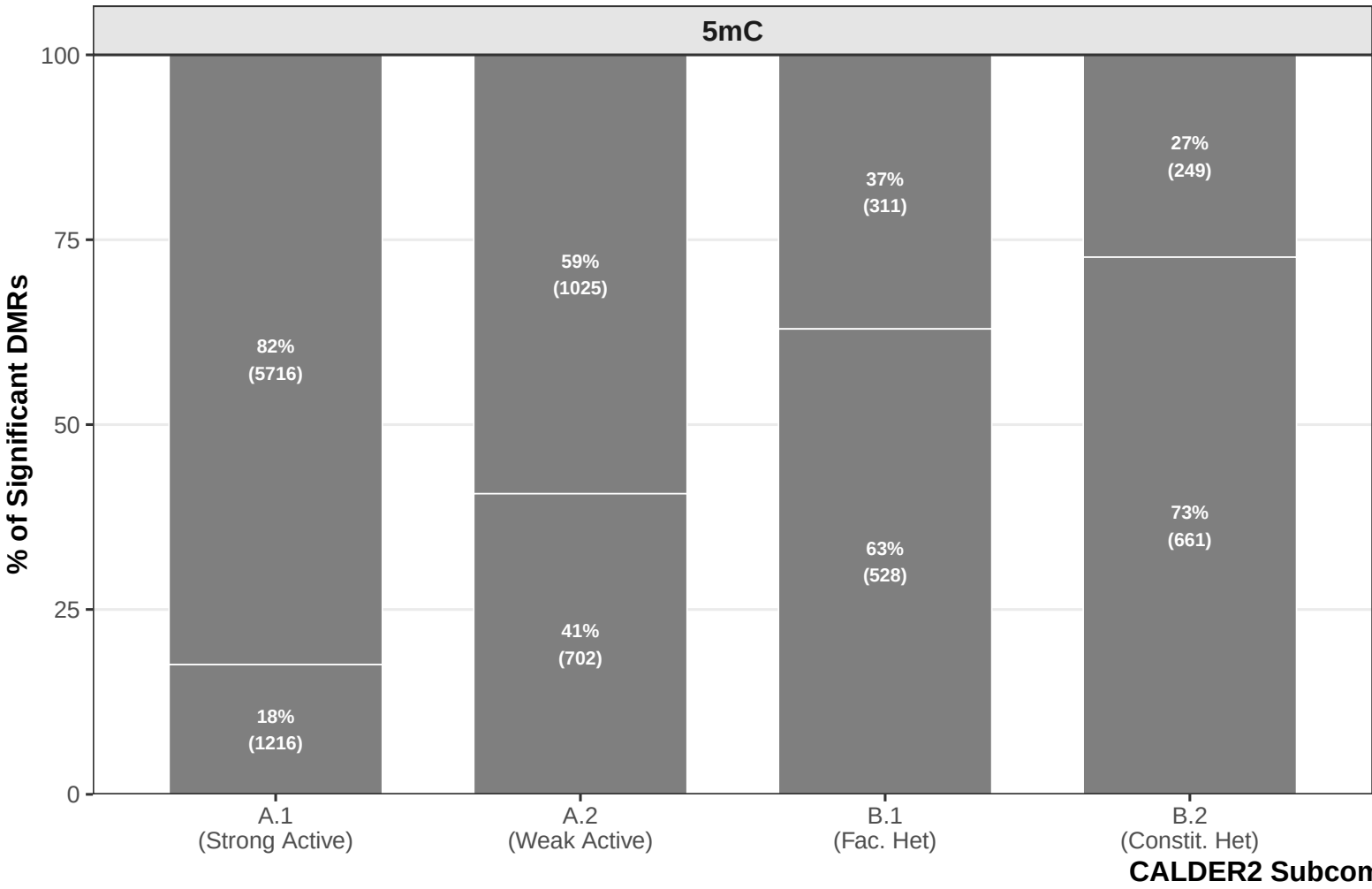
Compare directionality with 5mC (Section 66b)

Direction ■ Hypermethylated ■ Hypomethylated



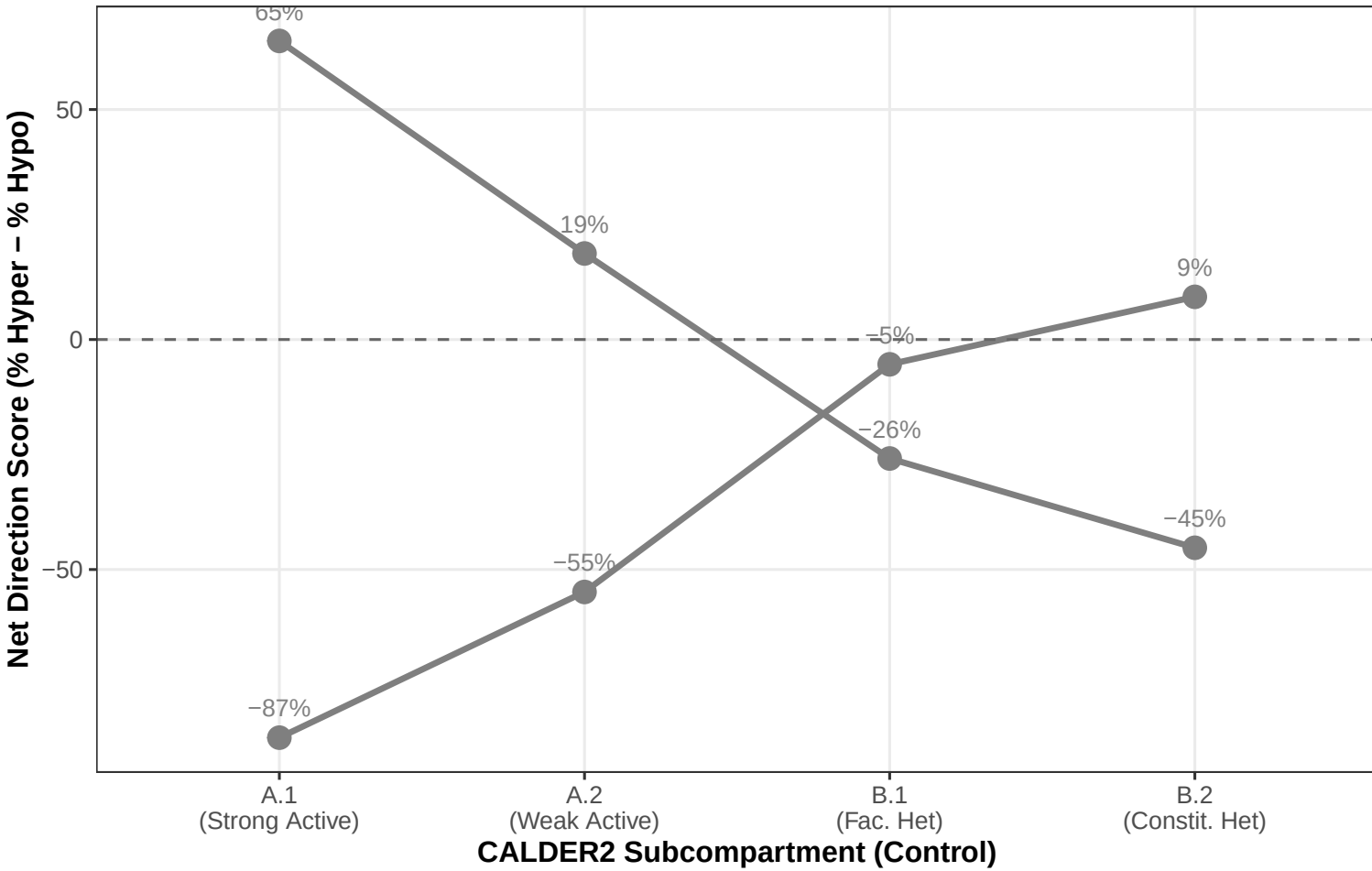
5mC vs 5hmC Direction Split by Subcompartment (Side-by-Side)

Active chromatin (A.1): 5mC gains methylation, 5hmC loses it – mirror-image gradients



Methylation Gradient: Net Direction (% Hyper – % Hypo) by Subcompartment

Which modality shows a steeper gradient from active to heterochromatic?



Median Methylation Change by Subcompartment

Absolute effect sizes across the compartment spectrum

